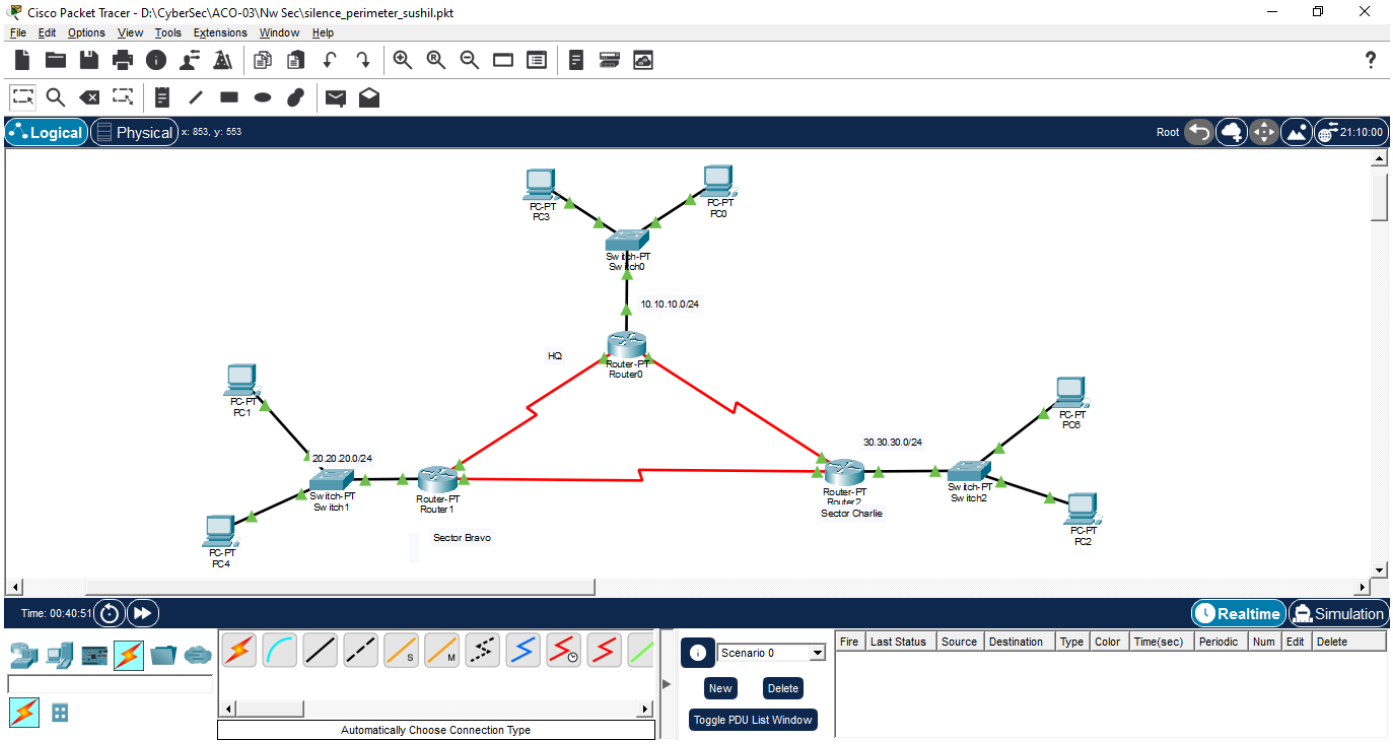
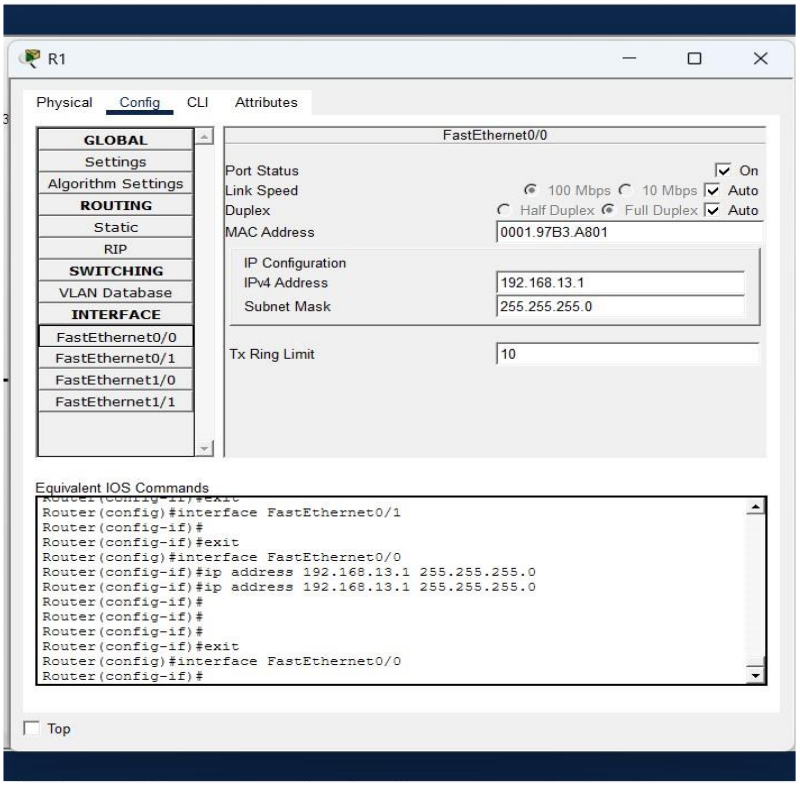
Phase 1 : Network Restoration

1. Network Topology



Fault Introduced : Wrong Wildcard Introduced



Step 1: Connection check in IP route(not seen)

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.10.0/24 is directly connected, FastEthernet1/0
L       10.10.10.2/32 is directly connected, FastEthernet1/0
     20.0.0.0/24 is subnetted, 1 subnets
O       20.20.20.0/24 [110/2] via 192.168.12.2, 00:20:32, FastEthernet0/1
     30.0.0.0/24 is subnetted, 1 subnets
O E2    30.30.30.0/24 [110/20] via 192.168.12.2, 00:20:32, FastEthernet0/1
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/30 is directly connected, FastEthernet0/1
L       192.168.12.1/32 is directly connected, FastEthernet0/1
     192.168.13.0/24 is variably subnetted, 3 subnets, 3 masks
C       192.168.13.0/24 is directly connected, FastEthernet0/0
O E2    192.168.13.0/30 [110/20] via 192.168.12.2, 00:01:38, FastEthernet0/1
L       192.168.13.1/32 is directly connected, FastEthernet0/0
     192.168.23.0/30 is subnetted, 1 subnets
O E2    192.168.23.0/30 [110/20] via 192.168.12.2, 00:20:32, FastEthernet0/1

Router#
```

Copy Paste

Step 2 : Ping Successful after rectification

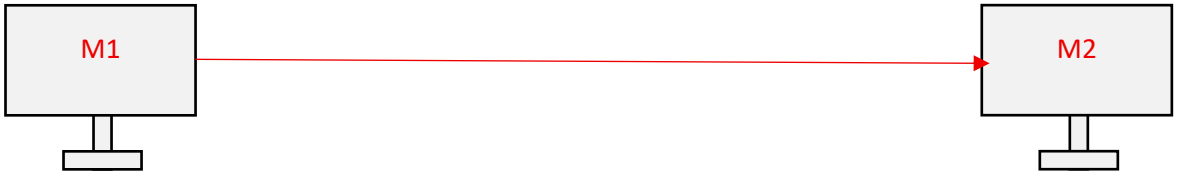
R1

Physical Config CLI Attributes

IOS Command Line Interface

```
     20.0.0.0/24 is subnetted, 1 subnets
O       20.20.20.0/24 [110/2] via 192.168.12.2, 00:20:32, FastEthernet0/1
     30.0.0.0/24 is subnetted, 1 subnets
O E2    30.30.30.0/24 [110/20] via 192.168.12.2, 00:20:32, FastEthernet0/1
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/30 is directly connected, FastEthernet0/1
L       192.168.12.1/32 is directly connected, FastEthernet0/1
     192.168.13.0/24 is variably subnetted, 3 subnets, 3 masks
C       192.168.13.0/24 is directly connected, FastEthernet0/0
O E2    192.168.13.0/30 [110/20] via 192.168.12.2, 00:01:38, FastEthernet0/1
L       192.168.13.1/32 is directly connected, FastEthernet0/0
     192.168.23.0/30 is subnetted, 1 subnets
O E2    192.168.23.0/30 [110/20] via 192.168.12.2, 00:20:32, FastEthernet0/1

Router#
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.13.1 255.255.255.252
Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 30.30.30.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 30.30.30.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Phase-2 Intercept Credentials



Machine-01 Data

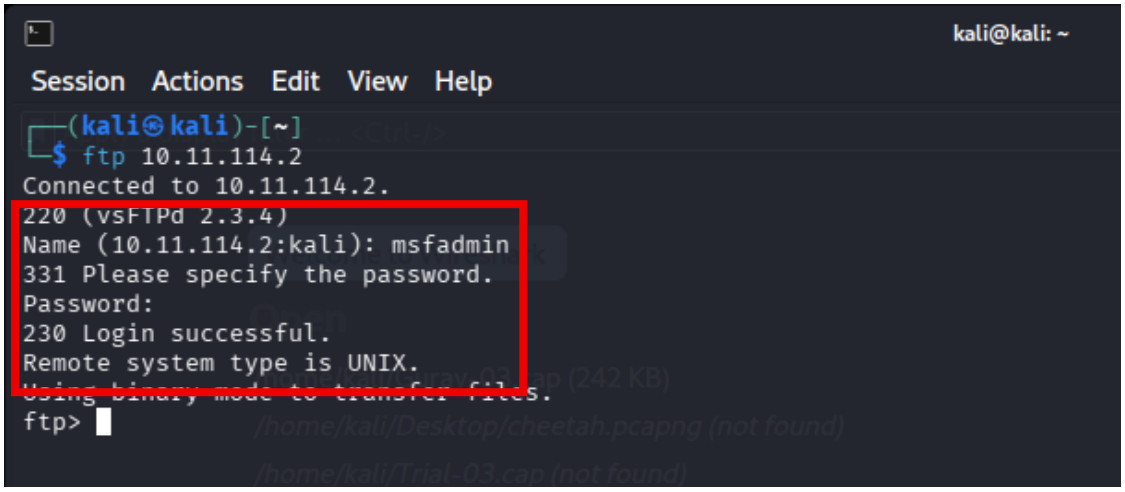
Role: Attacker
OS: Kali Linux
IP : 10.11.114.72

Machine-02 Data

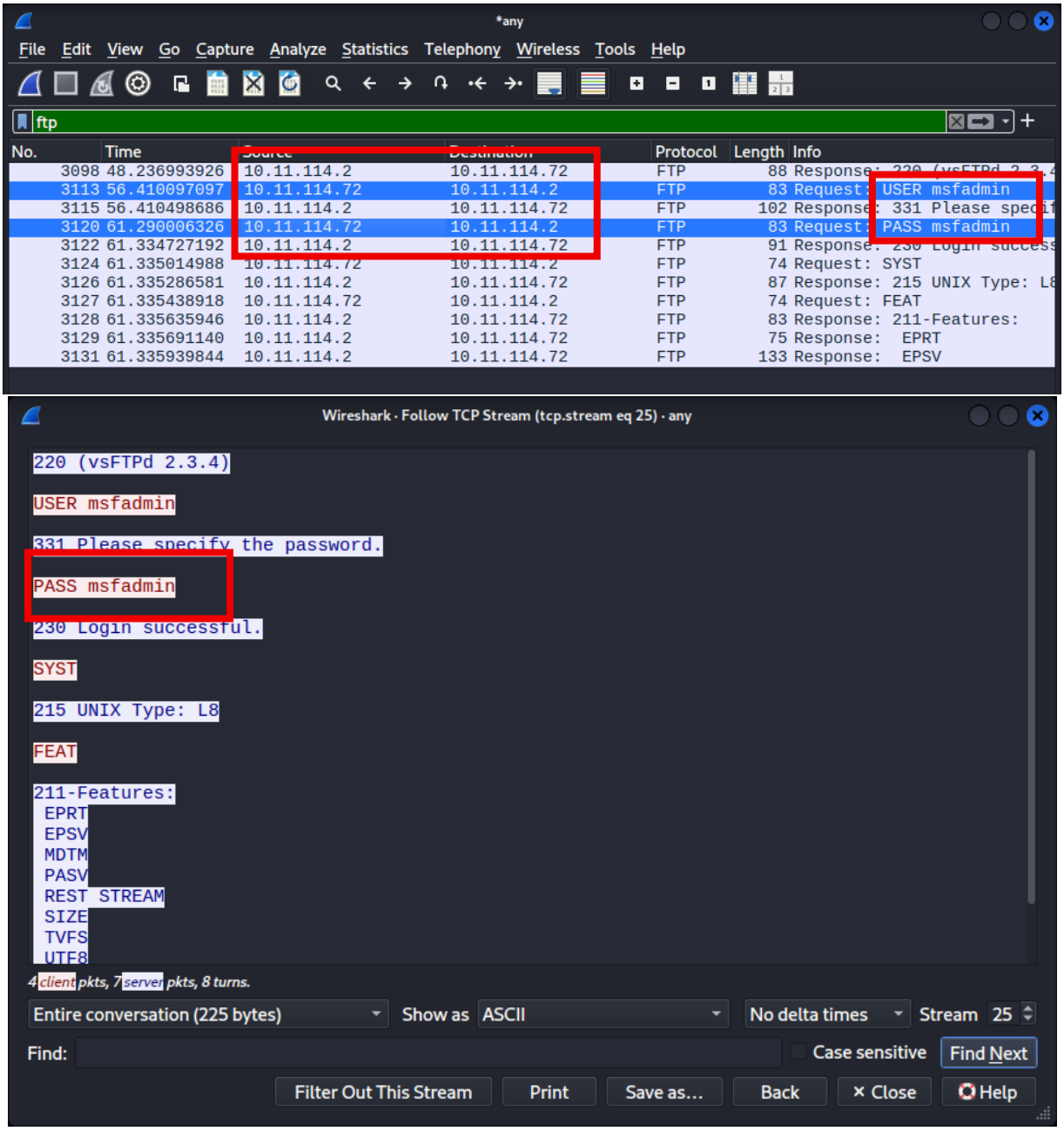
Role: Victim
OS: Ubuntu(Metaspolitable 2)
IP : 10.11.114.2

SCENARIO 1 : FTP

Step 1: Host Scanning

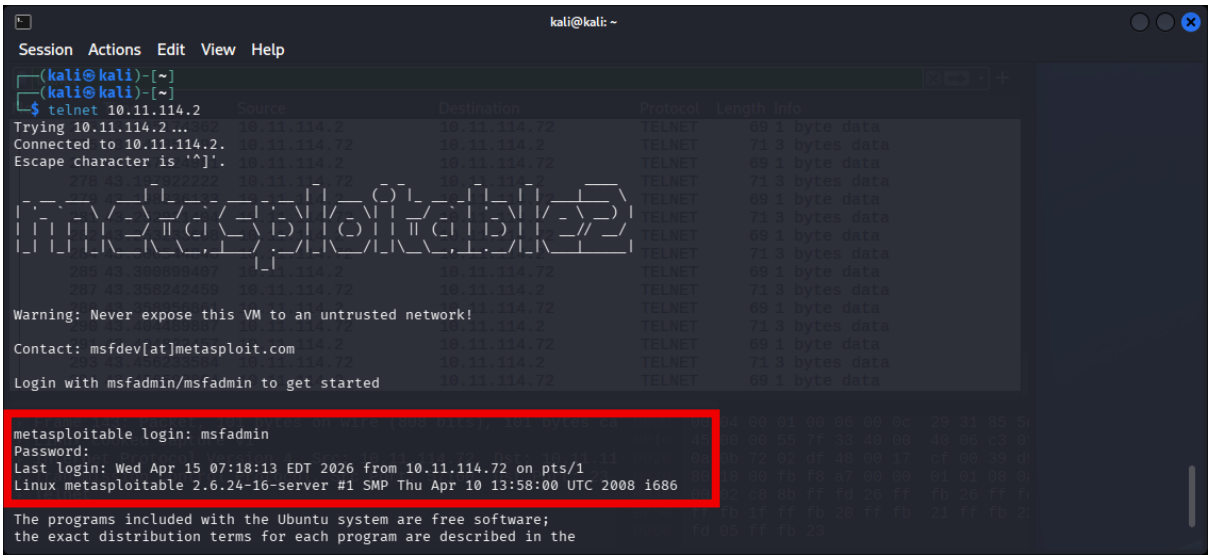


Step 3: Capture Details in Wireshark



SCENARIO 2 : TELNET

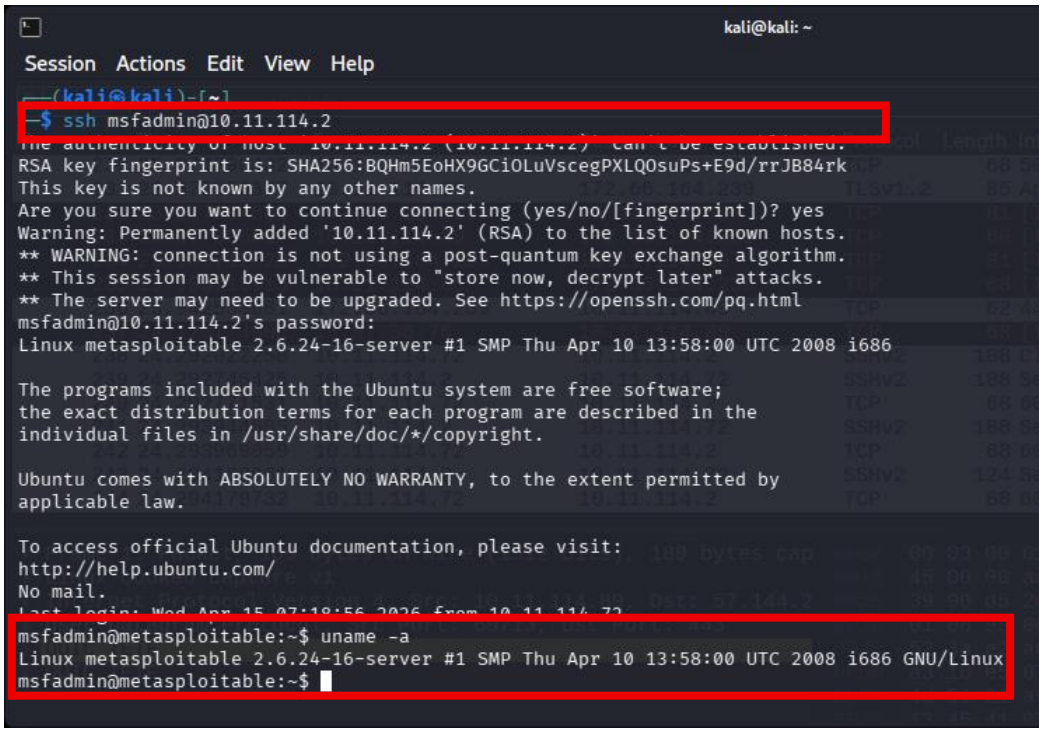
Step 1 : Telnet Login



Step 2 : Packet Capture

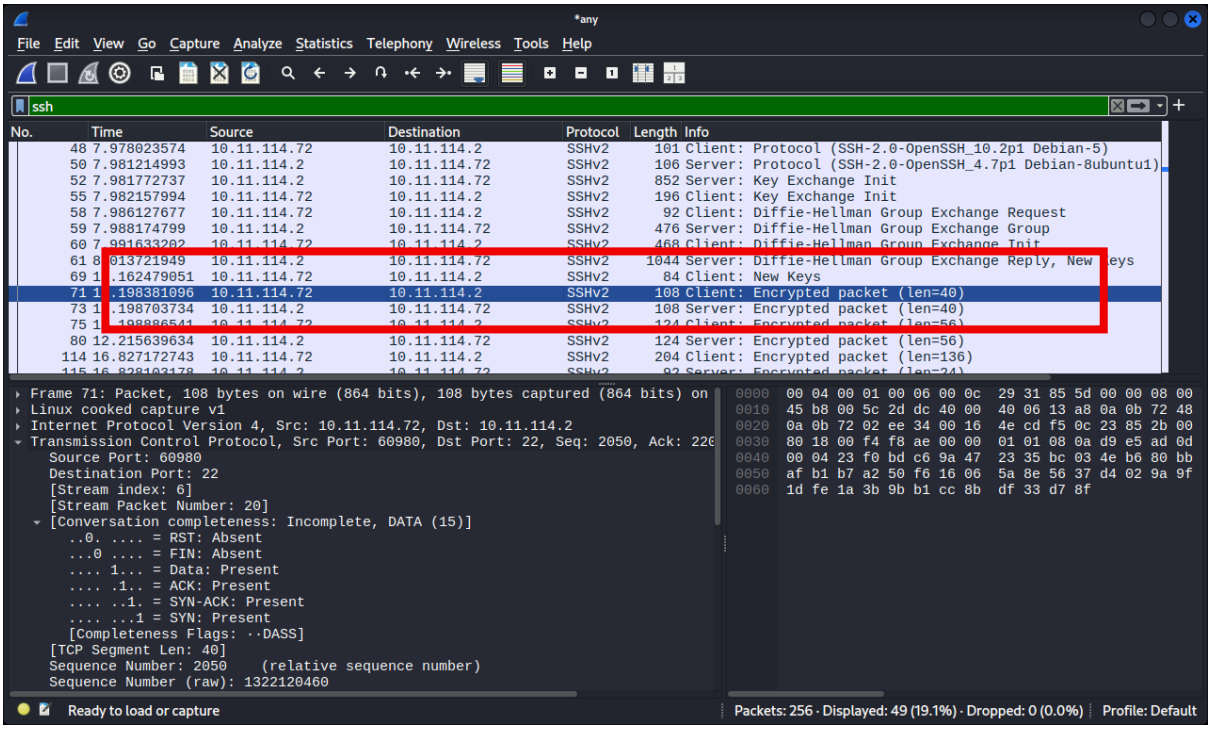
SCENARIO 3 : SSH

Step 1 : SSH Login

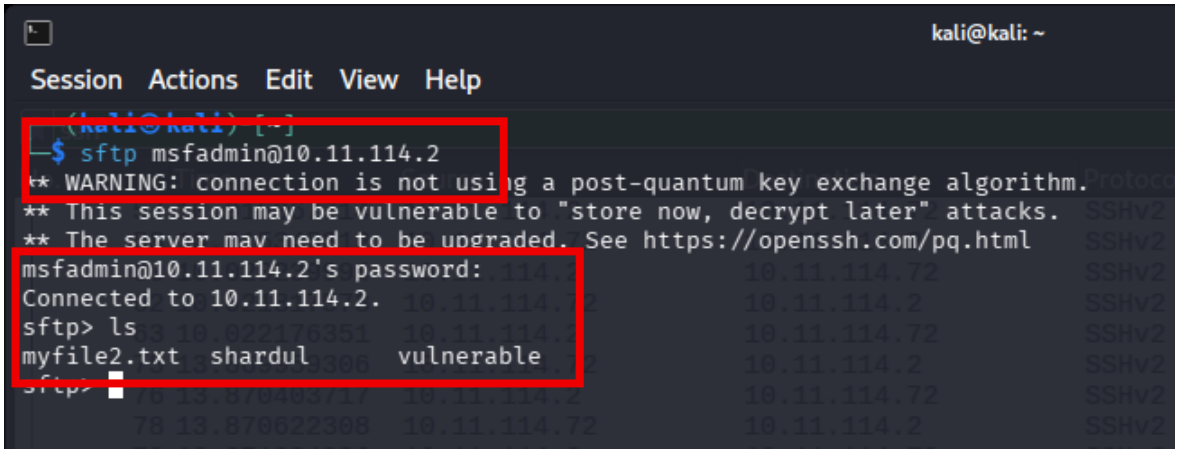


Step 2 : Packet Capture

Encrypted Data. Username & Password not Captured in plain

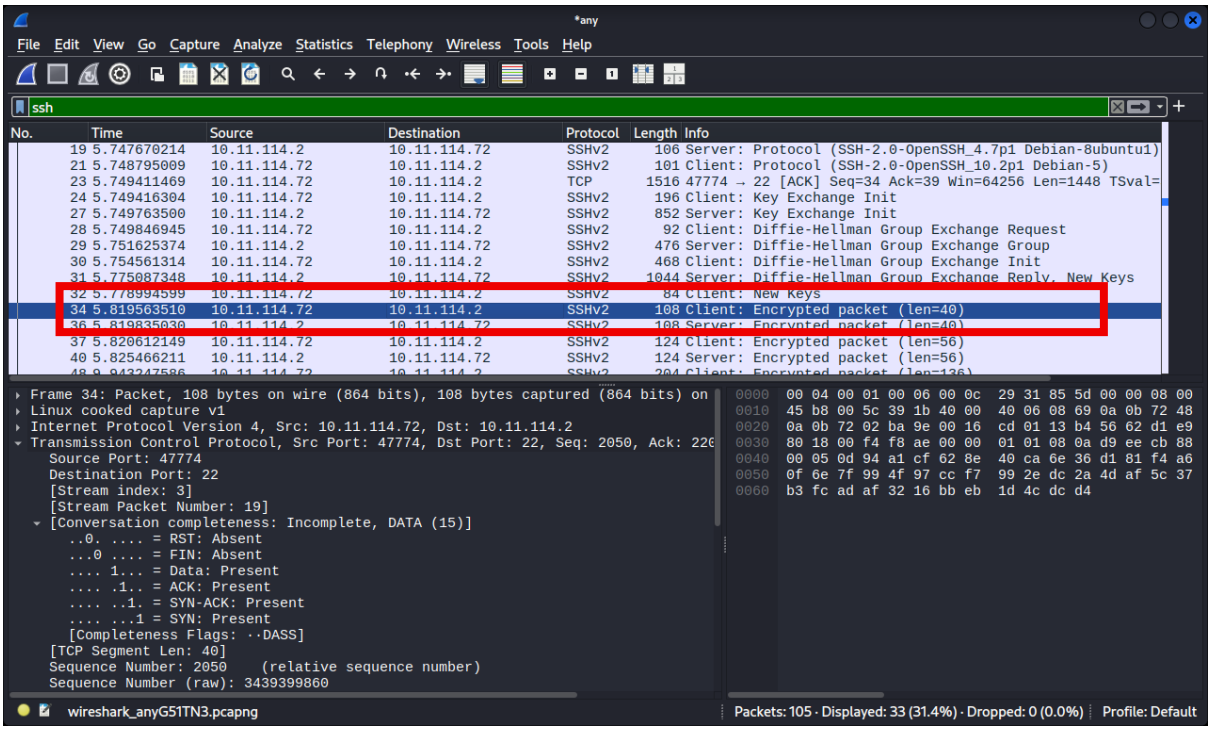


SCENARIO 4 : SFTP



Step 2 : Packet Capture

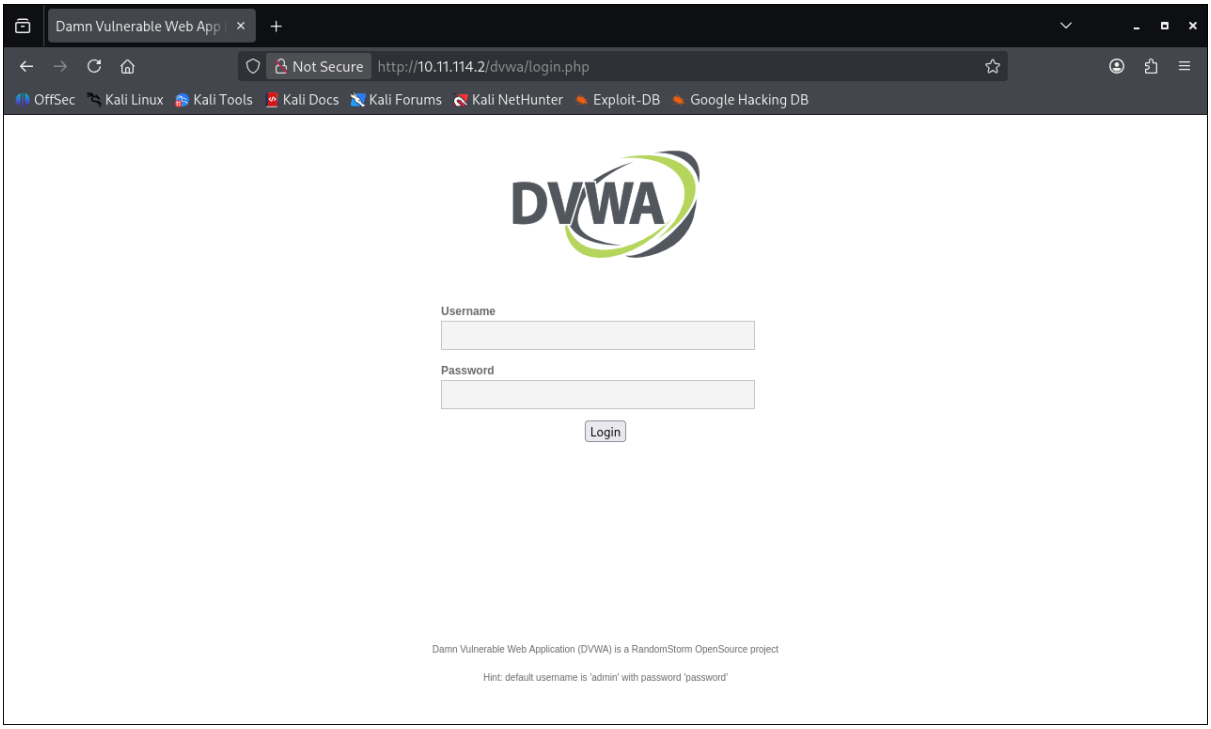
Encrypted Data. Username & Password not Captured in plain



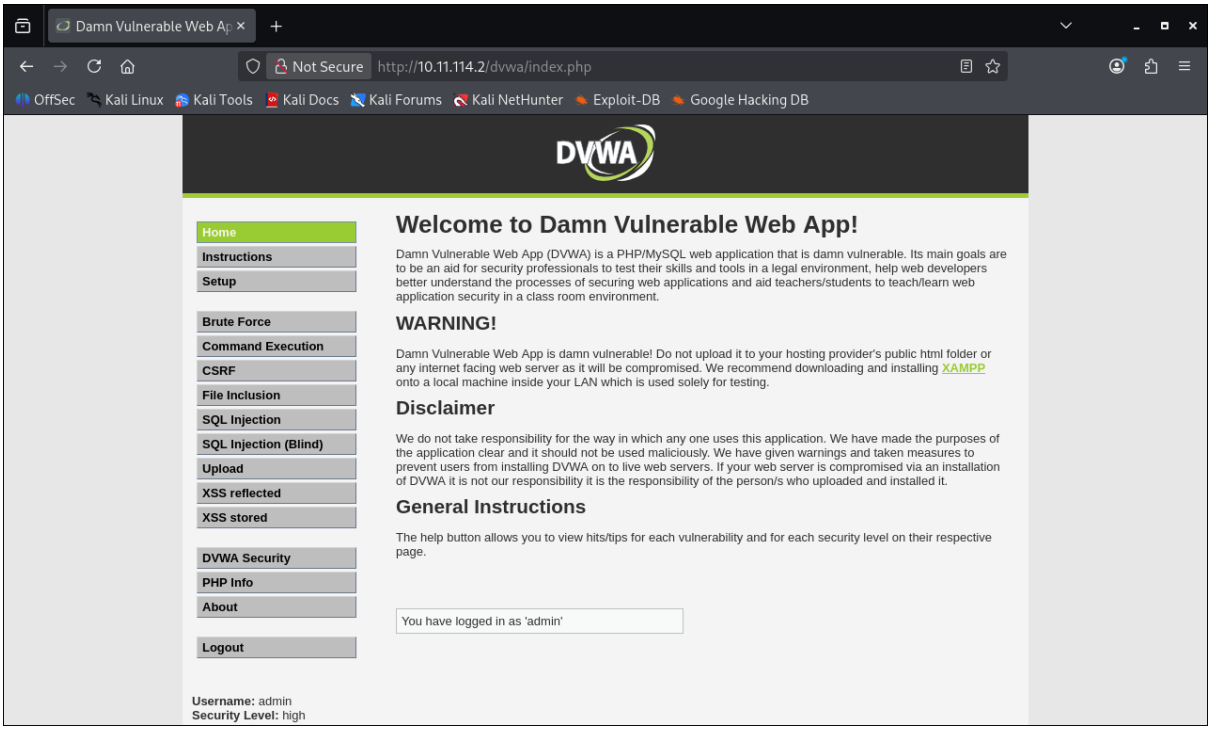
This is the reason why encryption is required. FTP sends credentials in plaintext whereas SFTP sends credentials in encrypted form. Thus a MITM or a network tfc sniffer cannot directly access credentials just by capturing packets.

SCENARIO 5: DVWA

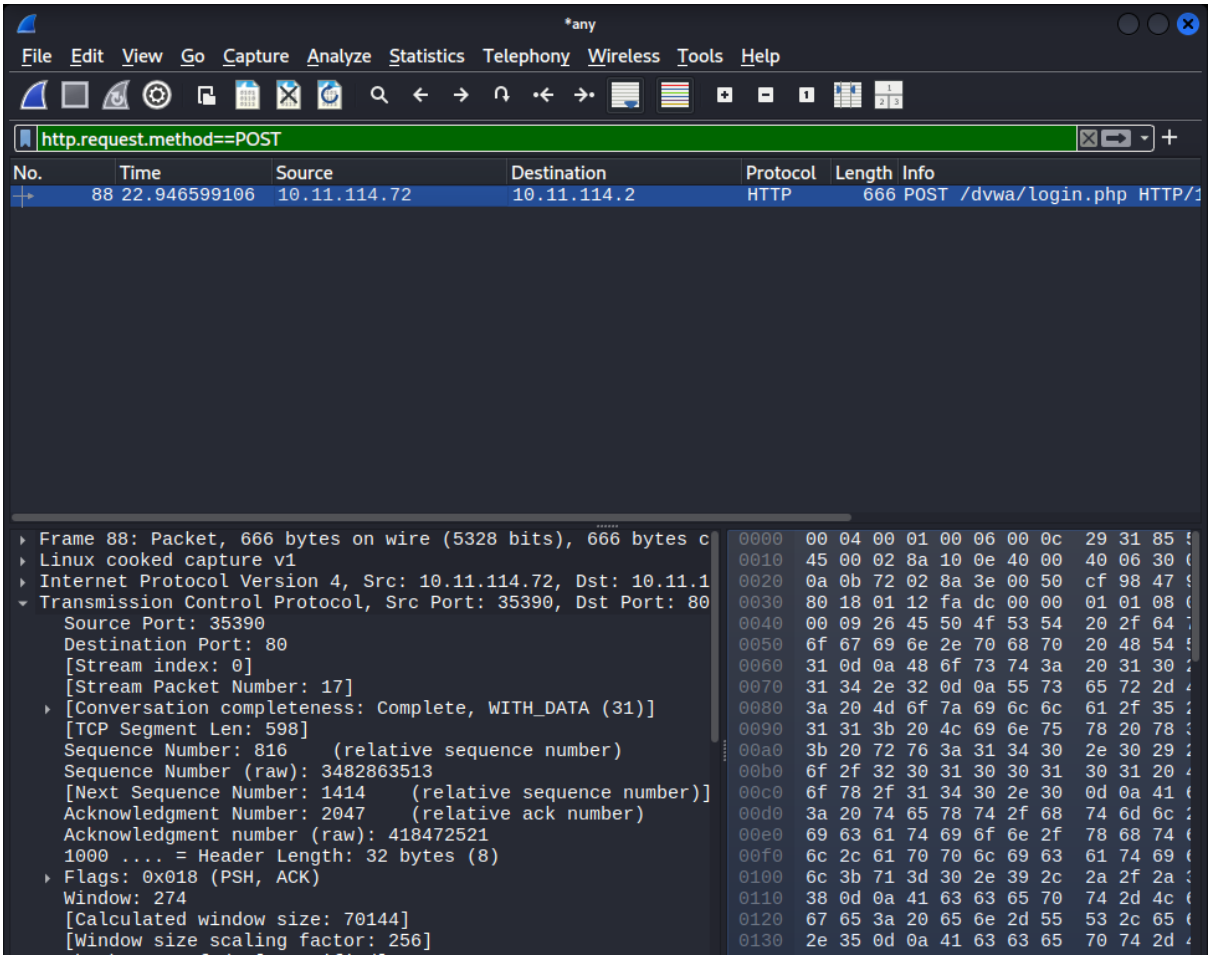
Step 1: Access DVWA Portal running on Metaspolitable 2 and login



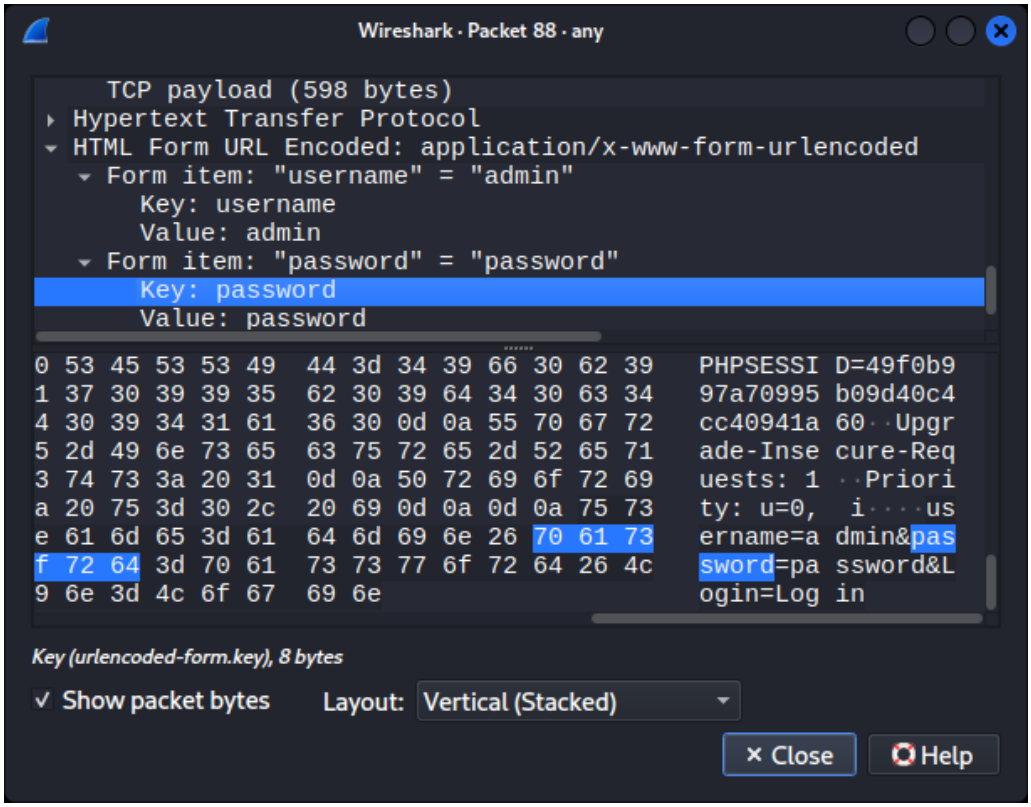
Step 2 : Login Successful



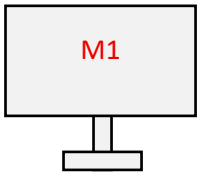
Step 3 : Packet Capture

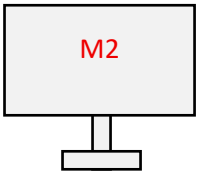


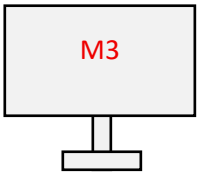
Step 4 : Credentials Captured



Phase 3: Intrusion Detection Battle(Suricata)







Machine-01 Data

OS: Metasploitable 2
IP : 192.168.1.7

Machine-02 Data

OS: Windows(VM)
IP : 192.168.1.5

Machine-03 Data

OS: Kali Linux
IP : 192.168.1.8

Step 1: Ping Detection

```
kali@kali: /var/log/suricata
Session  Actions  Edit  View  Help
04/15/2026-21:54:07.198292  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:5bbf:5fd6:38dc:9e1d:131 → ff02:0000:0000:0000:0000:0000:0000:00fb:0

┌──(kali㉿kali)-[/var/log/suricata]
└─$ tail fast.log
04/15/2026-21:54:02.708337  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:5bbf:5fd6:38dc:9e1d:131 → ff02:0000:0000:0000:0000:0001:ffdc:9e1d:0
04/15/2026-21:54:04.151667  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:52ca:dfb2:bd2d:58e6:131 → ff02:0000:0000:0000:0000:0001:ff2d:58e6:0
04/15/2026-21:54:05.710783  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:5bbf:5fd6:38dc:9e1d:131 → ff02:0000:0000:0000:0000:0000:0000:000c:0
04/15/2026-21:54:05.754764  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:e0d5:3f47:71ee:a861:131 → ff02:0000:0000:0000:0000:0000:0001:0003:0
04/15/2026-21:54:06.206333  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:5bbf:5fd6:38dc:9e1d:131 → ff02:0000:0000:0000:0000:0000:0001:0003:0
04/15/2026-21:54:06.827794  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:020c:29ff:fe31:855d:131 → ff02:0000:0000:0000:0000:0001:ff31:855d:0
04/15/2026-21:54:07.198292  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:5bbf:5fd6:38dc:9e1d:131 → ff02:0000:0000:0000:0000:0000:0000:00fb:0
04/15/2026-21:55:44.309793  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.1.3:
8 → 192.168.1.8:0
04/15/2026-21:55:44.309827  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.1.8:
0 → 192.168.1.3:0
04/15/2026-21:55:47.370728  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:0000:0000:0000:0001:131 → ff02:0000:0000:0000:0000:0001:ff00:0001:0

┌──(kali㉿kali)-[/var/log/suricata]
└─$
```

Step 2: SYN Scan

```
kali@kali: ~
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ nmap -sS 192.168.1.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 22:41 +0530
Nmap scan report for 192.168.1.5
Host is up (0.00042s latency).
All 1000 scanned ports on 192.168.1.5 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:1E:6F:83 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 22.81 seconds

┌──(kali㉿kali)-[~]
└─$ tail /var/log/suricata/fast.log
04/15/2026-22:42:16.771685  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:4249
7 → 192.168.1.5:1007
04/15/2026-22:42:16.774229  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:4249
7 → 192.168.1.5:42
04/15/2026-22:42:16.774364  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:4249
7 → 192.168.1.5:9011
04/15/2026-22:42:16.798777  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:4249
9 → 192.168.1.5:8654
04/15/2026-22:42:16.822983  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:4249
9 → 192.168.1.5:1434
04/15/2026-22:42:16.823130  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:4249
9 → 192.168.1.5:1600
04/15/2026-22:42:16.853626  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:4249
9 → 192.168.1.5:8873
```

Step 3 : Port Scanning

```
kali@kali: ~
Session  Actions  Edit  View  Help
04/15/2026-23:03:28.964712  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:5675
2 → 192.168.1.5:80

┌──(kali㉿kali)-[~]
└─$ tail -10 /var/log/suricata/fast.log
04/15/2026-23:03:24.964405  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:5675
2 → 192.168.1.5:21
04/15/2026-23:03:26.964424  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:5675
04/15/2026-23:03:26.964424  [**] [1:2000006:1] Custom:Port Scanning in Progress [**] [Classification: (null)] [Priority: 3] {TCP} 192.168
.1.8:56750 → 192.168.1.5:80
2 → 192.168.1.5:80
04/15/2026-23:03:30.965167  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:5675
0 → 192.168.1.5:23
04/15/2026-23:03:30.965167  [**] [1:4000001:1] Custom: Telnet Attempt [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:56750
 → 192.168.1.5:23
04/15/2026-23:03:32.964278  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:5675
2 → 192.168.1.5:23
04/15/2026-23:03:32.964278  [**] [1:4000001:1] Custom: Telnet Attempt [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:56752
 → 192.168.1.5:23
04/15/2026-23:03:34.965058  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:5675
0 → 192.168.1.5:20
04/15/2026-23:03:36.964517  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:5675
2 → 192.168.1.5:20

┌──(kali㉿kali)-[~]
└─$
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
└─$ nmap -p 20,21,23,80,443 --max-rate 0.5 192.168.1.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 23:03 +0530
Nmap scan report for 192.168.1.5
Host is up (0.00047s latency).

PORT    STATE    SERVICE
20/tcp  filtered ftp-data
21/tcp  filtered ftp
23/tcp  filtered telnet
80/tcp  filtered http
443/tcp filtered https
MAC Address: 00:0C:29:1E:6F:83 (VMware)

┌──(kali㉿kali)-[~]
└─$
```

Step 4 : Telnet

```
kali@kali: ~
Session  Actions  Edit  View  Help
04/15/2026-23:12:06.428968  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:0000:0000:0000:0001:134 → ff02:0000:0000:0000:0000:0000:0000:0001:0

┌──(kali㉿kali)-[~]
└─$ tail -10 /var/log/suricata/fast.log
04/15/2026-23:11:20.413815  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:5bbf:5fd6:38dc:9e1d:131 → ff02:0000:0000:0000:0000:0000:0000:00fb:0
04/15/2026-23:11:20.413816  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:5bbf:5fd6:38dc:9e1d:131 → ff02:0000:0000:0000:0000:0000:0000:000c:0
04/15/2026-23:11:22.540866  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:52ca:dfb2:bd2d:58e6:131 → ff02:0000:0000:0000:0000:0000:0000:000c:0
04/15/2026-23:11:23.539905  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:52ca:dfb2:bd2d:58e6:131 → ff02:0000:0000:0000:0000:0000:0000:00fb:0
04/15/2026-23:11:25.540352  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:52ca:dfb2:bd2d:58e6:131 → ff02:0000:0000:0000:0000:0001:ff2d:58e6:0
04/15/2026-23:11:25.544120  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:52ca:dfb2:bd2d:58e6:131 → ff02:0000:0000:0000:0000:0000:0001:0003:0
04/15/2026-23:12:04.940286  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:5965
0 → 192.168.1.7:23
04/15/2026-23:12:04.940286  [**] [1:4000001:1] Custom: Telnet Attempt [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:59650
 → 192.168.1.7:23
04/15/2026-23:12:06.421114  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:020c:29ff:fe31:855d:133 → ff02:0000:0000:0000:0000:0000:0000:0002:0
04/15/2026-23:12:06.428968  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:0000:0000:0000:0001:134 → ff02:0000:0000:0000:0000:0000:0000:0001:0

┌──(kali㉿kali)-[~]
└─$
```

Step 5 : FTP

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ftp 192.168.1.7
Connected to 192.168.1.7.
220 (vsFTPd 2.3.4)
Name (192.168.1.7:kali): msfadmin
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
Nmap done: 1 IP address (1 host up) scanned in 20.66 seconds

┌──(kali㉿kali)-[~]
└─$ tail /var/log/suricata/fast.log
04/15/2026-23:13:24.411530  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:5bbf:5fd6:38dc:9e1d:131 -> ff02:0000:0000:0000:0000:0000:0001:0003:0
04/15/2026-23:13:24.909990  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:020c:29ff:fe9c:9899:131 -> ff02:0000:0000:0000:0000:0001:ff9c:9899:0
04/15/2026-23:13:25.404443  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:5bbf:5fd6:38dc:9e1d:131 -> ff02:0000:0000:0000:0000:0000:0000:00fb:0
04/15/2026-23:13:29.041575  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:52ca:dfb2:bd2d:58e6:131 -> ff02:0000:0000:0000:0000:0000:00fb:0
04/15/2026-23:13:29.541508  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:52ca:dfb2:bd2d:58e6:131 -> ff02:0000:0000:0000:0000:0000:0000:000c:0
04/15/2026-23:13:32.531910  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:52ca:dfb2:bd2d:58e6:131 -> ff02:0000:0000:0000:0000:0000:0001:0003:0
04/15/2026-23:13:34.538186  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:52ca:dfb2:bd2d:58e6:131 -> ff02:0000:0000:0000:0000:0001:ff2d:58e6:0
04/15/2026-23:14:01.503099  [**] [1:6000001:1] Custom:FTP User/Pass Credential Exposure [**] [Classification: (null)] [Priority: 3] {TCP}
 192.168.1.7:21 -> 192.168.1.8:59290
04/15/2026-23:14:07.042401  [**] [1:300001:2] Custom:SYN Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.1.8:5750
2 -> 192.168.1.7:21
04/15/2026-23:15:13.554817  [**] [1:1000001:1] Custom: ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:00
00:0000:0000:0000:0000:0000:0001:131 -> ff02:0000:0000:0000:0000:0001:ff00:0001:0

┌──(kali㉿kali)-[~]
└─$
```

Phase 4: Host Forensics

1. Netstat -an

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ netstat -an
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 192.168.1.8:57502       192.168.1.7:21          ESTABLISHED
tcp6       0      0 :::80                   :::*                    LISTEN
udp        0      0 192.168.1.8:68          192.168.1.1:67          ESTABLISHED
udp6       0      0 fe80::20c:29ff:fe31:546 :::*
raw6       0      0 :::58                   :::*                    7
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     21178
unix  3      [ ]         STREAM     CONNECTED     19848    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     19110
unix  3      [ ]         STREAM     CONNECTED     21574
unix  3      [ ]         STREAM     CONNECTED     18800
unix  3      [ ]         STREAM     CONNECTED     18386
unix  3      [ ]         STREAM     CONNECTED     17831
unix  3      [ ]         STREAM     CONNECTED     16843    /run/user/1000/at-spi/bus_0
unix  2      [ ]         DGRAM      CONNECTED     9773
unix  3      [ ]         STREAM     CONNECTED     78434
unix  3      [ ]         STREAM     CONNECTED     16548    /run/systemd/journal/stdout
unix  2      [ ]         DGRAM      CONNECTED     16131
unix  3      [ ]         STREAM     CONNECTED     16639
unix  3      [ ]         STREAM     CONNECTED     12623
unix  2      [ ACC ]     STREAM     LISTENING     11356    /run/systemd/io.systemd.sysext
unix  3      [ ]         STREAM     CONNECTED     13643
unix  3      [ ]         STREAM     CONNECTED     54707
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
unix  3      [ ]         STREAM     CONNECTED     12623
unix  2      [ ACC ]     STREAM     LISTENING     11356    /run/systemd/io.systemd.sysext
unix  3      [ ]         STREAM     CONNECTED     13643
unix  3      [ ]         STREAM     CONNECTED     54707
unix  3      [ ]         STREAM     CONNECTED     19928    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     17491
unix  3      [ ]         STREAM     CONNECTED     19080    /tmp/.ICE-unix/1243
unix  3      [ ]         STREAM     CONNECTED     16813    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     21008    /run/systemd/journal/stdout
unix  3      [ ]         DGRAM      CONNECTED     16170
unix  3      [ ]         STREAM     CONNECTED     14960
unix  3      [ ]         STREAM     CONNECTED     21539    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     20904
unix  3      [ ]         STREAM     CONNECTED     19921
unix  3      [ ]         STREAM     CONNECTED     17515    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     15779
unix  3      [ ]         STREAM     CONNECTED     20021    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     54703
unix  3      [ ]         STREAM     CONNECTED     19121    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     20829    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     19749
unix  3      [ ]         STREAM     CONNECTED     17731    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     17447
unix  3      [ ]         STREAM     CONNECTED     16817    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     12472    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     16600
unix  2      [ ACC ]     STREAM     LISTENING     13779    /tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     20942
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
unix  3      [ ]         STREAM     CONNECTED     12623
unix  2      [ ACC ]     STREAM     LISTENING     11356    /run/systemd/io.systemd.sysext
unix  3      [ ]         STREAM     CONNECTED     13643
unix  3      [ ]         STREAM     CONNECTED     54707
unix  3      [ ]         STREAM     CONNECTED     19928    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     17491
unix  3      [ ]         STREAM     CONNECTED     19080    /tmp/.ICE-unix/1243
unix  3      [ ]         STREAM     CONNECTED     16813    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     21008    /run/systemd/journal/stdout
unix  3      [ ]         DGRAM      CONNECTED     16170
unix  3      [ ]         STREAM     CONNECTED     14960
unix  3      [ ]         STREAM     CONNECTED     21539    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     20904
unix  3      [ ]         STREAM     CONNECTED     19921
unix  3      [ ]         STREAM     CONNECTED     17515    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     15779
unix  3      [ ]         STREAM     CONNECTED     20021    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     54703
unix  3      [ ]         STREAM     CONNECTED     19121    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     20829    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     19749
unix  3      [ ]         STREAM     CONNECTED     17731    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     17447
unix  3      [ ]         STREAM     CONNECTED     16817    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     12472    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     16600
unix  2      [ ACC ]     STREAM     LISTENING     13779    /tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     20942
```

2. Netstat -ano

```
kali@kali: ~
Session  Actions  Edit  View  Help
unix  3      [ ]         STREAM     CONNECTED     12805    @1167ee472401515c/bus/systemd-logind/system

┌──(kali㉿kali)-[~]
└─$ netstat -ano
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       Timer
tcp        0      1 192.168.1.8:57502       192.168.1.7:21          FIN_WAIT1   on (0.00/4/0)
tcp6       0      0 :::80                   :::*                    LISTEN      off (0.00/0/0)
udp        0      0 192.168.1.8:68          192.168.1.1:67          ESTABLISHED off (0.00/0/0)
udp6       0      0 fe80::20c:29ff:fe31:546 :::*                                off (0.00/0/0)
raw6       0      0 :::58                   :::*                    7           off (0.00/0/0)
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     21178
unix  3      [ ]         STREAM     CONNECTED     19848    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     19110
unix  3      [ ]         STREAM     CONNECTED     21574
unix  3      [ ]         STREAM     CONNECTED     18800
unix  3      [ ]         STREAM     CONNECTED     18386
unix  3      [ ]         STREAM     CONNECTED     17831
unix  3      [ ]         STREAM     CONNECTED     16843    /run/user/1000/at-spi/bus_0
unix  2      [ ]         DGRAM      CONNECTED     9773
unix  3      [ ]         STREAM     CONNECTED     78434
unix  3      [ ]         STREAM     CONNECTED     16548    /run/systemd/journal/stdout
unix  2      [ ]         DGRAM      CONNECTED     16131
unix  3      [ ]         STREAM     CONNECTED     16639
unix  3      [ ]         STREAM     CONNECTED     12623
unix  2      [ ACC ]     STREAM     LISTENING     11356    /run/systemd/io.systemd.sysext
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
unix  3      [ ]         STREAM     CONNECTED     13607    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     13139
unix  3      [ ]         STREAM     CONNECTED     21007
unix  3      [ ]         STREAM     CONNECTED     20948
unix  3      [ ]         STREAM     CONNECTED     19735
unix  3      [ ]         STREAM     CONNECTED     21548    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     17942
unix  3      [ ]         STREAM     CONNECTED     88495
unix  3      [ ]         STREAM     CONNECTED     17459    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     17713
unix  2      [ ACC ]     STREAM     LISTENING     16911    /tmp/.ICE-unix/1243
unix  3      [ ]         STREAM     CONNECTED     17840
unix  3      [ ]         DGRAM      CONNECTED     9150
unix  2      [ ]         DGRAM      CONNECTED     89126
unix  3      [ ]         STREAM     CONNECTED     19907
unix  3      [ ]         STREAM     CONNECTED     17484    @/tmp/.X11-unix/X0
unix  3      [ ]         DGRAM      CONNECTED     16165
unix  3      [ ]         STREAM     CONNECTED     21488
unix  3      [ ]         STREAM     CONNECTED     16659    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     16537    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     21172
unix  3      [ ]         STREAM     CONNECTED     21073
unix  3      [ ]         STREAM     CONNECTED     20825
unix  3      [ ]         STREAM     CONNECTED     16439    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     17830    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     18269    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     21553
unix  3      [ ]         STREAM     CONNECTED     19803
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
unix  3      [ ]         DGRAM      CONNECTED     16169
unix  3      [ ]         STREAM     CONNECTED     9691
unix  3      [ ]         STREAM     CONNECTED     19908    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     16847    /run/user/1000/bus
unix  2      [ ]         DGRAM                    16164    /run/user/1000/systemd/notify
unix  3      [ ]         STREAM     CONNECTED     19770
unix  2      [ ACC ]     STREAM     LISTENING     16171    /run/user/1000/systemd/private
unix  3      [ ]         STREAM     CONNECTED     54702    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     18811    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     18316    /run/systemd/journal/stdout
unix  2      [ ACC ]     STREAM     LISTENING     16175    /run/user/1000/systemd/io.systemd.Manager
unix  2      [ ACC ]     STREAM     LISTENING     16186    /run/user/1000/bus
unix  3      [ ]         DGRAM      CONNECTED     16168
unix  3      [ ]         STREAM     CONNECTED     19927
unix  3      [ ]         STREAM     CONNECTED     17804    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     17506    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     16432
unix  3      [ ]         STREAM     CONNECTED     16045
unix  3      [ ]         STREAM     CONNECTED     17933    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     16514    /run/user/1000/bus
unix  2      [ ACC ]     STREAM     LISTENING     16190    /run/user/1000/gnupg/S.dirmngr
unix  3      [ ]         DGRAM      CONNECTED     9151
unix  3      [ ]         STREAM     CONNECTED     20835    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     78429
unix  3      [ ]         STREAM     CONNECTED     19757    @/tmp/.X11-unix/X0
unix  2      [ ACC ]     STREAM     LISTENING     16191    /run/user/1000/keyring/control
unix  3      [ ]         STREAM     CONNECTED     13047    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     18549    /run/user/1000/pulse/native
```

3. Netstat -rn

```
kali@kali: ~
Session  Actions  Edit  View  Help
unix  3      [ ]         STREAM     CONNECTED     12471
unix  2      [ ACC ]     STREAM     LISTENING     9422     /run/systemd/journal/io.systemd.journal
unix  3      [ ]         STREAM     CONNECTED     21176    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     17312
unix  3      [ ]         STREAM     CONNECTED     89358
unix  3      [ ]         STREAM     CONNECTED     21551    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     17971
unix  3      [ ]         STREAM     CONNECTED     13309
unix  2      [ ]         DGRAM      CONNECTED     9900
unix  2      [ ]         DGRAM                    9755     @14841585101318388229
unix  2      [ ACC ]     STREAM     LISTENING     16910    @/tmp/.ICE-unix/1243
unix  2      [ ACC ]     STREAM     LISTENING     13778    @/tmp/.X11-unix/X0
unix  2      [ ]         STREAM                   20865    @printer-applet-lock-user-kali
unix  3      [ ]         STREAM     CONNECTED     16435    @12929858be904ad7/bus/systemd/bus-api-user
unix  3      [ ]         STREAM     CONNECTED     13060    @9e7967e970ccce20/bus/systemd/bus-api-system
unix  3      [ ]         STREAM     CONNECTED     16172    @3a30ecbaf8a2a5b8/bus/systemd/bus-system
unix  3      [ ]         STREAM     CONNECTED     12346    @a24f2662ee6d5df6/bus/systemd-timesyn/bus-api-timesync
unix  3      [ ]         STREAM     CONNECTED     12805    @1167ee472401515c/bus/systemd-logind/system

┌──(kali㉿kali)-[~]
└─$ netstat -rn
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
0.0.0.0         192.168.1.1     0.0.0.0         UG        0 0          0 eth0
192.168.1.0     0.0.0.0         255.255.255.0   U         0 0          0 eth0

┌──(kali㉿kali)-[~]
└─$
```

4. Netstat -tulpn

```
kali@kali: ~
Session  Actions  Edit  View  Help
unix  2      [ ]         DGRAM                    9755     @14841585101318388229
unix  2      [ ACC ]     STREAM     LISTENING     16910    @/tmp/.ICE-unix/1243
unix  2      [ ACC ]     STREAM     LISTENING     13778    @/tmp/.X11-unix/X0
unix  2      [ ]         STREAM                   20865    @printer-applet-lock-user-kali
unix  3      [ ]         STREAM     CONNECTED     16435    @12929858be904ad7/bus/systemd/bus-api-user
unix  3      [ ]         STREAM     CONNECTED     13060    @9e7967e970ccce20/bus/systemd/bus-api-system
unix  3      [ ]         STREAM     CONNECTED     16172    @3a30ecbaf8a2a5b8/bus/systemd/bus-system
unix  3      [ ]         STREAM     CONNECTED     12346    @a24f2662ee6d5df6/bus/systemd-timesyn/bus-api-timesync
unix  3      [ ]         STREAM     CONNECTED     12805    @1167ee472401515c/bus/systemd-logind/system

┌──(kali㉿kali)-[~]
└─$ netstat -rn
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
0.0.0.0         192.168.1.1     0.0.0.0         UG        0 0          0 eth0
192.168.1.0     0.0.0.0         255.255.255.0   U         0 0          0 eth0

┌──(kali㉿kali)-[~]
└─$ netstat -tulpn
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp6       0      0 :::80                   :::*                    LISTEN      -
udp6       0      0 fe80::20c:29ff:fe31:546 :::*                                -

┌──(kali㉿kali)-[~]
└─$
```

5. Containment Steps

a. Iden the process ID which has est unauthorized comn
b. Kill the process using pskill/kill/taskkill
c. Block inbound/outbound tfc to a suspicious foreign IP, using firewall

Phase 5 : INCIDENT RESPONSE REPORT

1. **Initial Event**. Network Reconnaissance (Nmap Scan)
    1.1. The Nmap SYN scan indicates the attacker was enumerating open ports and services.
    .2. This is typically the first stage of the cyber kill chain (reconnaissance phase).
    1.3. **Conclusion**. The attack began with port scanning to identify vulnerable services
2. **Evidence**. Port Scan Capture
    2.1. Foreign IP(10.11.114.72) scanning a host(10.11.114.2)
    2.2. Packet analysis (via Wireshark logs) shows:
        2.2.1. username: msfadmin
        2.2.2. password: msfadmin
3. **Attacker Identification**
    3.1. Source IP observed in Nmap Scan logs
    3.2. Indicators.
        3.2.1. Repeated connections from same IP
        3.2.2. Port scanning behavior
        3.2.3. Unauthorized service enumeration
    3.3. Conclusion. The attacker is the external IP performing scanning + unauthorized access attempts

4. **Multiple Control Failures**.
    4.1. Lack of Encryption.Telnet and FTP used → plaintext exposure
    4.2. Weak IDS/IPS Response. Suricata detected activity but did not block it
    4.3. Conclusion. Failure of preventive controls (encryption, segmentation, firewall rules)
5. **Immediate Containment Actions**
    5.1. Isolate Affected Host. Disconnect laptop from network
    5.2. Block Attacker IP. Apply firewall rule: deny ip 10.11.114.72
    5.3. Disable Vulnerable Services. Stop Telnet and FTP
    5.4. Kill Suspicious Process.
    5.5. Enable Monitoring. Start packet capture and monitor logs in real-time
6. **Long-term remediation plan**
    6.1. Security Improvements
    6.1.1. Replace insecure protocols (Telnet, FTP)
    6.1.2. Enforce strong authentication (MFA)
    6.2. Network Hardening. Implement VLAN segmentation and Zero Trust model
    6.3. Detection & Prevention.

6.3.1 Upgrade IDS → IPS mode

6.3.2. Integrate logs with SIEM (e.g., Wazuh)

6.4. Endpoint Security.

6.4.1. Install EDR solutions

6.4.2. Monitor unusual ports

6.5. Training. User awareness on credential security